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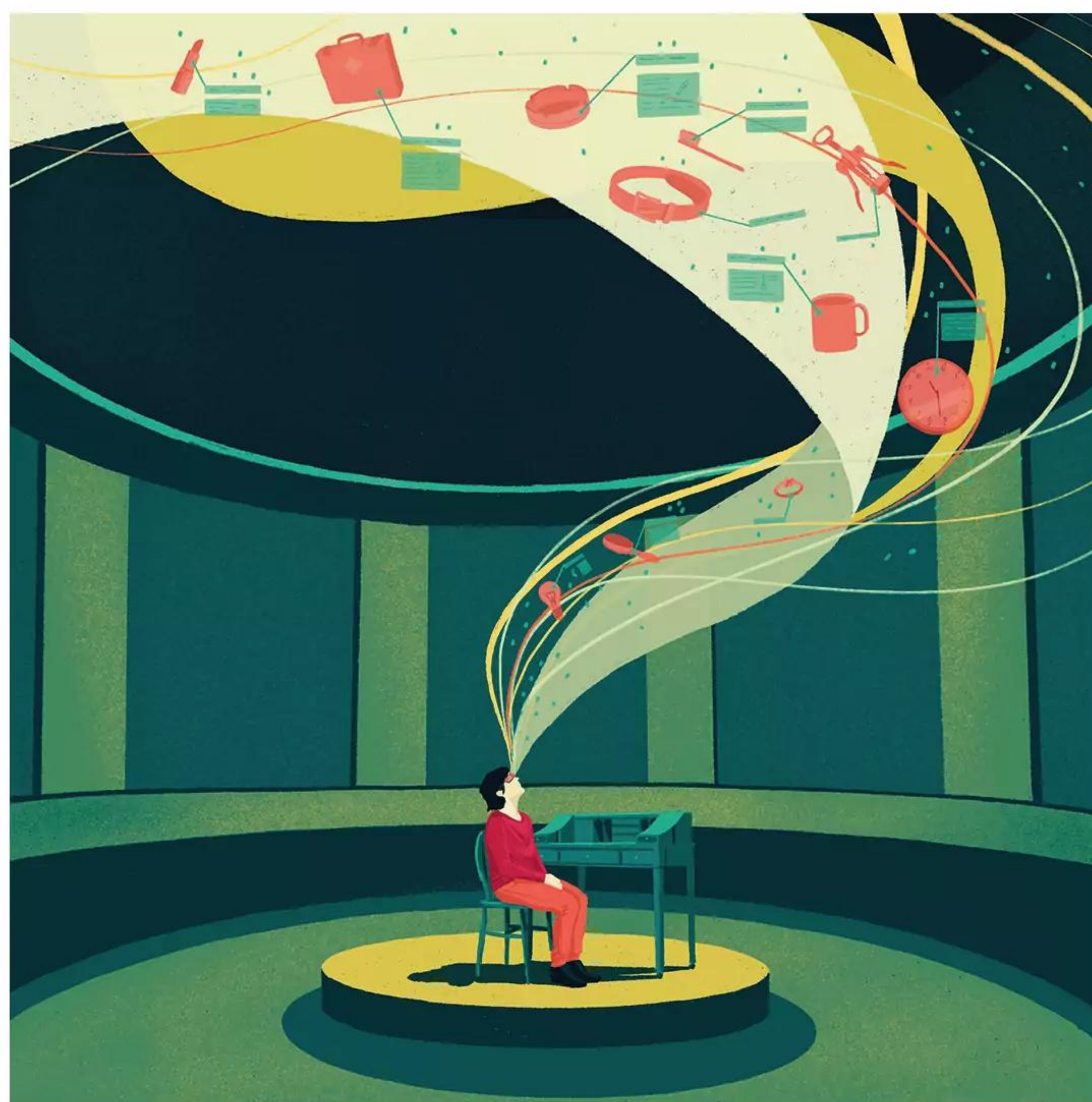
New technologies will radically change the experience of living with and caring for someone with Alzheimer's, says **Fiona Carragher**

THE past few years have seen outstanding progress in medication for previously incurable conditions. Most pertinent for me is the approval of lecanemab for use in the UK and elsewhere to slow the progress of early-stage Alzheimer's disease. It isn't a cure, but it brings us closer to dementia becoming a chronic condition managed with drugs.

But while lecanemab has been hailed as a game changer, it isn't yet readily available and, in the meantime, 55 million people are living with dementia globally. By 2050, that will grow to 139 million. It is one of the biggest medical and social challenges facing the planet and needs innovative and radical solutions – like new technologies.

We are increasingly interacting with artificial intelligence in daily life as our personal devices have become powerful assistants, offering predictive spelling in texts and face recognition in photo apps. These everyday technologies can be leveraged to help those with dementia, nudging people on daily routines and helping them feel independent at home.

For the past decade, we have seen bulky wearables and crude monitoring systems, used by families desperate to help their loved ones live at home for as long as possible, while still being concerned about their safety. These tools had a vital role, but modern AI-driven tech has far outpaced their ability, both in accurate reporting and in upholding the dignity of people with dementia.



SIMONE ROTELLA

Creating a new wave of helpful technologies requires careful work with medical and tech experts, in-depth testing and co-creation with those who have experience of the condition. People with dementia need purpose-designed systems that can adapt as their condition progresses, rather than remain static. This is exactly what 24 innovators have been working on over the past year, competing to become finalists in the Longitude Prize on Dementia.

The scheme is co-funded by the British government's Innovate UK agency and the charity Alzheimer's Society, where

I work, and delivered by the social enterprise Challenge Works (part of Nesta). It is driving the development of AI and machine learning-based tech to help people with dementia live independently. From the 24 semi-finalists, we have five finalist technologies, revealed this week.

These innovations truly hold the potential to revolutionise dementia care. Animorph Co-operative has developed high-tech glasses that help users recognise objects and people. The lenses, which work with eyesight prescriptions, display a halo around the object in view and text

that labels it ("mug") and describes what to do with it ("drink the tea"). Clairvoyant Networks has adapted ultra-wideband technology – a high-precision sensor previously used to track footballs during the World Cup – to create an AI system that predicts and prevents falls in those with dementia by monitoring indicators, such as gait, via wearable tech.

The other finalists include groups producing a smartwatch that can nudge you on routines, a traditional telephone that reminds you of activities and a camera-free, at-home monitoring box that detects where someone is in their house and updates carers on any concerning changes in routine while protecting user privacy. All five will be awarded funding to further develop their products and, in 2026, one will be named winner.

New treatments like lecanemab won't be suitable or available for everyone, so these technologies have huge potential. They will provide a critical mid-step for people with early-stage dementia to help them lead confident, fulfilled lives for longer, radically changing the experience of living with and caring for someone with the condition. This will herald a new era for dementia care and I, for one, think this can't come soon enough. ■



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